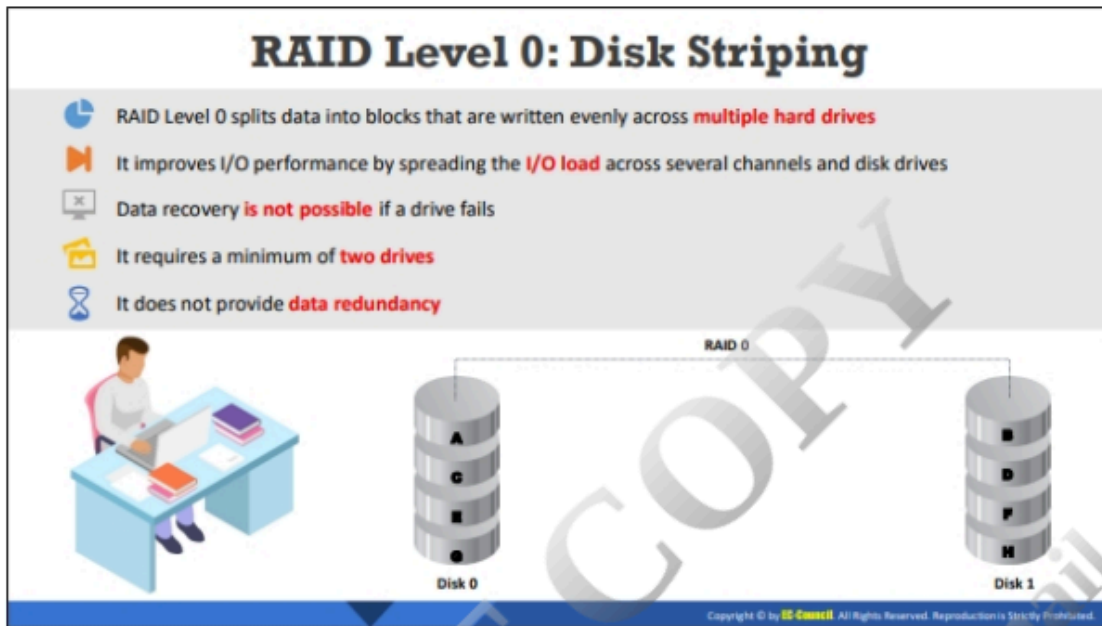




RAID Level 0: Disk Striping

Depending on the requirement of your organization, you can choose any RAID level. RAID levels are based on performance, fault tolerance, or both.

RAID 0 deals with data performance. In this level, data is broken into sections and written across multiple drives. The storage capacity of RAID 0 is equal to the sum of the disks' capacities in the set. RAID 0 does not provide fault tolerance. It requires a minimum of two drives. It does not provide data redundancy. A failure of one disk can lead to the failure of all disks in a level 0 volume. The probability of recovering data from a RAID level 0 is minimal.

The data distribution in a RAID level 0 is equal among all the disk sets, resulting in high performance. With concurrent high performance, the throughput of the read and write operations on multiple disks is equal to the throughput of the array of disks. Increased throughput is an advantage of RAID 0, considering that data recovery is not available. Software and hardware RAID controllers support RAID 0, helping boost server performance.

Example: Assume that the IT infrastructure has a hard disk with high performance. The data in the hard disk is transferred at a remarkably high speed. All the large and critical files are stored in this disk. However, if this disk fails, the entire contents of the files will be affected, leading to the unavailability of the data. It is advisable not to store any critical data in a RAID level 0.

Advantages of RAID Level 0

- **Read and write performance:** RAID level 0 has a good read and write performance. The performance is even better when the controller supports independent reads and writes to different disks in the array.
- **Cost:** RAID level 0 is more cost effective than the other RAID levels.

- **Implementation:** Is easy to implement as the data is divided in a sequential set of blocks. There is no storage loss as the maximum capacity is utilized.

Disadvantages of RAID Level 0

- **No redundancy:** With no data redundancy, data loss is greater.
- **Noncritical data:** Data that is not critical to the organization can be stored on RAID level 0. This level does not use mirroring. Recovery is not possible if critical data is lost on RAID level 0.
- **Unreliable:** If one disk fails, the entire network will be affected.



Figure 15.87: RAID level 0



RAID Level 1: Disk Mirroring

A typical RAID 1 contains an exact copy of the data on two or more disks. RAID 1 writes data on multiple drives and multiple mirror drives at the same time. The failure of one drive does not affect the data on other drives. This allows data retrieval from the mirror drive. Similar to RAID 0, RAID 1 provides no parity, striping, or spanning of disk space across multiple disks. RAID 1 can be used in accounting, payroll, and other financial applications.

RAID 1 is suitable in environments where read performance matters more than write performance. RAID 1 has improved read performance because the data in a disk can be read at the same time simultaneously.

RAID level 1 provides data reliability in case of a disk failure as it can still provide access to the same data mirrored on other disks. In a RAID 1 hardware implementation, a minimum of two disks are required. In a software RAID 1, data can be copied to a disk volume. RAID 1 reduces the total disk capacity by half.

Example: If a RAID 1 server is configured with two 4 TB drives, the storage capacity will be 4 TB and not 8 TB.

The drive that accesses the data first will service the request. The write throughput in RAID 1 is always slower because every drive needs to be updated. Thus, its performance is limited by the slowest drive. It is only as fast as its slowest drive. RAID 1 will continue to function as long as there is at least one working drive.

Advantages of RAID Level 1

- **High read performance:** Because there are two disks, the read performance is higher in a RAID level 1 system. Data can be read simultaneously while being written on the other disk. Thus, the redundancy is excellent.

- **Compatibility:** RAID 1 is compatible with hardware and software RAID systems, including controllers.
- **Reliability:** The mirroring feature in a RAID 1 ensures the data will be available, making it more reliable than a RAID level 0.

Disadvantages of RAID Level 1

- **Capacity:** RAID level 1 undergoes duplexing, which needs twice the amount of disk space for storage.
- **Hot swapping unavailable:** If a disk fails to run, it cannot be replaced while the server is still in operation. This is called hot swapping. RAID level 1 does not support hot swapping.



Figure 15.88: RAID level 1



The diagram illustrates RAID Level 3, Disk Striping with Parity. On the left, a laptop screen shows a 'Recovery' progress bar at 25% with a 'CANCEL' button. The main title is 'RAID Level 3: Disk Striping with Parity'. Below the title, three numbered points explain the RAID level: 01 Data is striped at the **byte level** across multiple drives. One drive per set is taken up for parity information. 02 If a drive fails, **data recovery and error correction** is possible using the parity drive in the set. 03 The **parity drive** stores the information on multiple drives. To the right, a diagram shows five disks labeled Disk 0 through Disk 4. Data is striped across Disk 0, Disk 1, Disk 2, and Disk 3 at the byte level. Disk 4 is designated as the parity drive, labeled 'Ap', 'Bp', 'Cp', and 'Dp'. A 'Parity Generation' box is shown above Disk 4, with arrows indicating the flow of data from the other disks to the parity drive. A copyright notice at the bottom reads: 'Copyright © by EG-Council. All Rights Reserved. Reproduction is Strictly Prohibited.'

RAID Level 3: Disk Striping with Parity

RAID level 3 is disk striping with parity. It uses striping and parity as its main features to store data. To implement a RAID level 3 system, a minimum of three disks are required. The data is stored on multiple drives at the byte level. This RAID level dedicates one drive to store the parity information. The byte level division allows the drives to work simultaneously. At any point in time, either a read operation or a write operation can take place. RAID 3 is a good choice for specialized databases or single-user systems.

The RAID level 3 has a high transfer data rate along with data security. It can perform data recovery and error correction by calculating an exclusive OR (XOR) of the information recorded on the parity drive.

Advantages of RAID Level 3

- **High throughput:** RAID level 3 provides a high throughput for read and write operations for large data transfers.
- **Resistant:** This RAID level is resistant to disk failures and breakdowns.

Disadvantages of RAID Level 3

- **Complexity:** Installation and configuration of a RAID level 3 system is complex. Its implementation necessitates a larger number of resources.
- **Slow performance:** Random operations affect its performance, thereby reducing its speed.

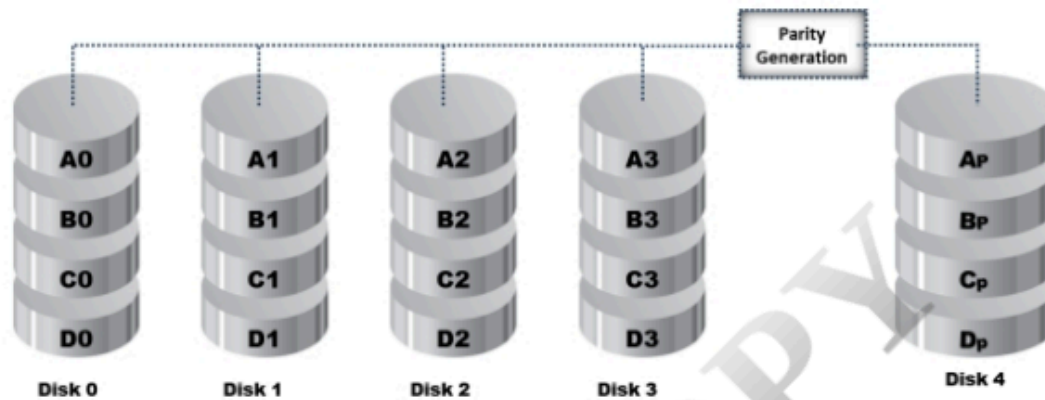
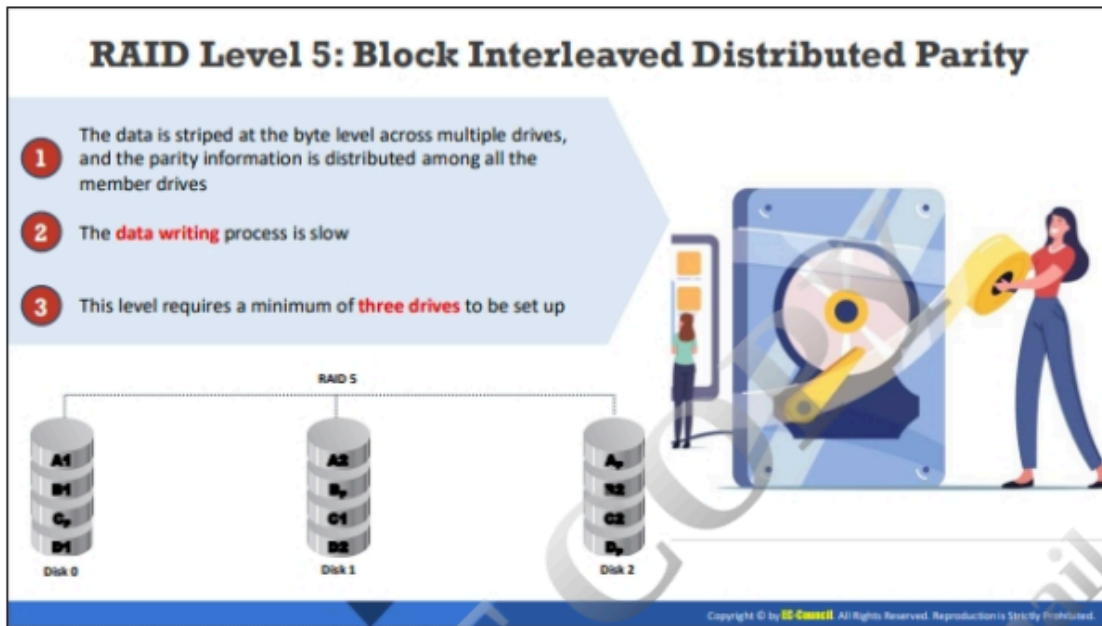


Figure 15.89: RAID level 3



RAID Level 5: Block Interleaved Distributed Parity

RAID level 5 involves a block-interleaved distributed parity; it includes a block-level striping with a distributed parity. The parity information is distributed among all drives except one. The data chunks in a RAID level 5 system are larger than the regular I/O size, but they can be resized. To prevent data loss following a drive failure, data can be calculated from the distributed parity.

The RAID 5 needs at least three disks; however, more than three disks can be used for a better performance. RAID 5 is not a good choice for write operations on the system. Rebuilding the RAID 5 array following a disk failure takes a long time. The performance can be degraded when the array is being rebuilt, making it vulnerable to additional disk failures. This level offers significantly better read performance as the disks independently process the data requests.

RAID 5 is most often found in file and application servers, database servers, along with web, email, and news servers.

Advantages of RAID Level 5

- **Read data:** Among all RAID levels, level 5 has the highest read data transaction rates.
- **Withstand failure:** RAID 5 can withstand the failure of a single drive and is not affected by the loss of data.
- **Hot swapping:** In case of a disk failure, the failed disk can be replaced with a new one, without a server shutdown.

Disadvantages of RAID Level 5

- **Slow write operation:** Servers built using RAID 5 suffer from performance issues with write operations, and these can eventually result in reduced speeds.

Example: Employees accessing a database on a RAID 5 server will reduce the server's production time.

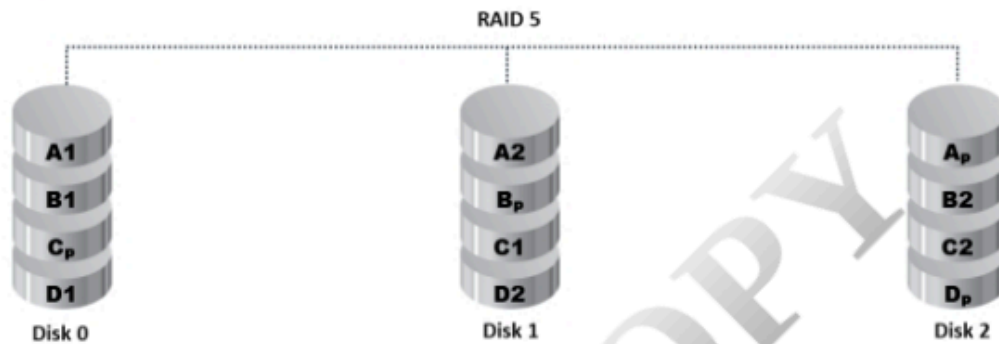
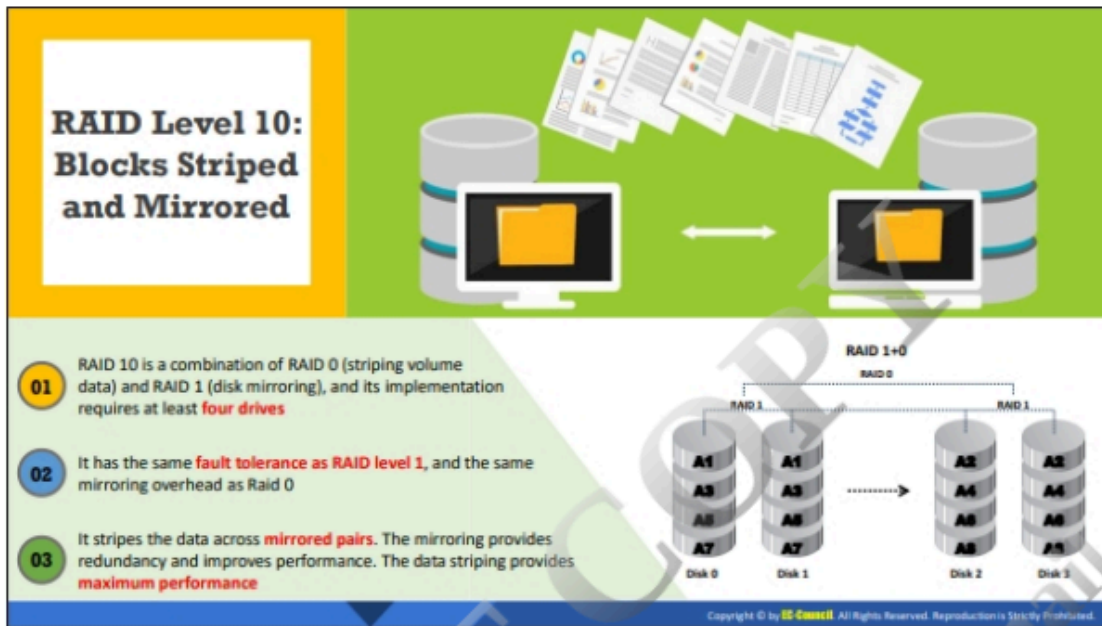


Figure 15.90: RAID level 5



RAID Level 10: Blocks Striped and Mirrored

RAID level 10 includes disk striping and mirroring in a nested hybrid RAID level. It is a combination of RAID level 1 and RAID level 0. It is also called as a “stripe of mirrors.” This level can symbolically be represented as RAID 1+0 or RAID 10. RAID 10 includes the mirroring of RAID 1 without parity and the striping of RAID 0. The performance of RAID 10 is higher than a RAID 1. RAID level 10 has the same fault tolerance as RAID level 1. It requires a minimum of four drives for its operation. RAID 10 is a great choice for database servers, web servers, email servers, etc., and can be implemented on hardware or software. The mirroring provides redundancy and improves performance. The data striping provides maximum performance.

Advantages of RAID Level 10

- **Improved I/O operations:** With a combination of RAID levels 1 and 0, it provides improved I/O operations.
- **Better throughput:** Compared with other RAID levels, RAID 10 provides better throughput and higher latency.
- **Efficient write operations:** The write operations are efficient in this level. Therefore, RAID 10 is often implemented on database servers and other servers that perform write operations.

Disadvantages of RAID Level 10

- **Expensive:** RAID 10 is expensive than other RAID levels as it requires twice as many disks.

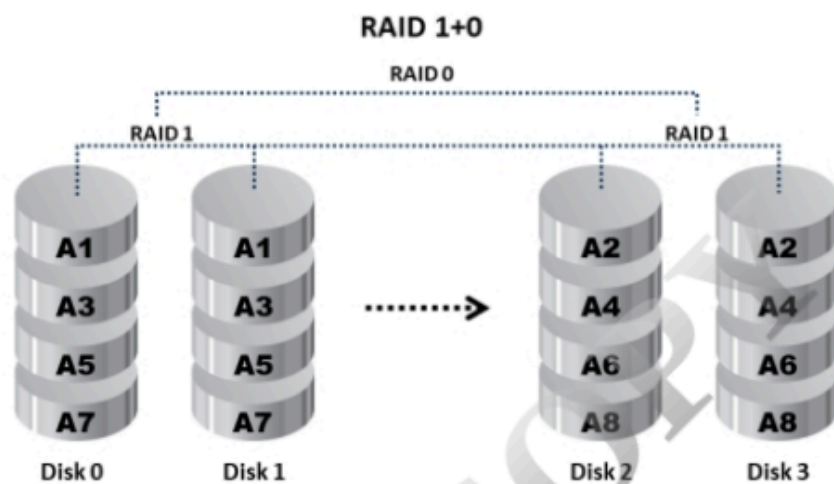
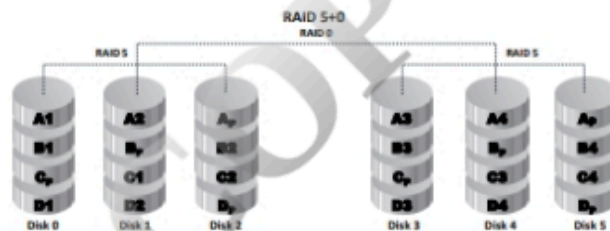


Figure 15.91: RAID level 10

RAID Level 50: Mirroring and Striping across Multiple RAID Levels



- ❑ RAID 50 is a combination of **RAID 0 striping** and the distributed parity of **RAID 5**
- ❑ It is **more fault tolerant** than a RAID 5 but uses twice the parity overhead
- ❑ A minimum of **six drives** are required for setup. A drive from each segment can fail and the array will recover. If more than one drive fails in a segment, the array will stop functioning.
- ❑ This RAID level offers better reads and writes than a RAID 5 and the highest levels of **redundancy** and **performance**



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RAID Level 50: Mirroring and Striping across Multiple RAID Levels

RAID level 50 includes mirroring and striping across multiple RAID levels. This level is a combination of a level 0 block-level striping and a level 5 distributed parity. The configuration of RAID level 50 requires a minimum of six drives. This level undergoes a hot swapping process when a disk fails. A drive from each segment can fail and the array will recover. If more than one drive fails in a segment, the array will stop functioning.

RAID 50 is an improvement over RAID 5, specifically for its write operation and fault tolerance. RAID level 50 can be implemented on servers that run applications requiring higher fault tolerance, capacity, and random access performance. This level offers data protection and faster rebuilds than a RAID 5 system. A failed disk in a segment only affects that segment and not the entire array. Only that segment is then rebuilt. The rest of the array functions normally.

Advantages

- **Security:** The data stored in a RAID 50 is more secured than in a RAID 5. With a larger storage capacity, this level offers better security than RAID 5.
- **Nondegradable:** With the use of a minimum of six drives in the configuration, failure of one disk does not impact the server function on this level.
- **Read and write performance:** The read and write performance of RAID level 50 is far better than RAID level 5.

Disadvantages

- **Controller:** Only a sophisticated controller can handle RAID level 50.

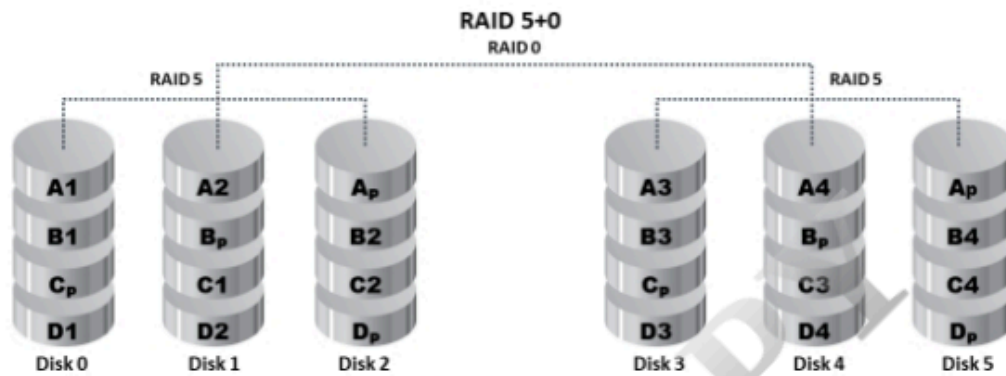


Figure 15.92: RAID level 50